

MAJOR PROJECT REPORT: EXECUTIVE VISUAL INTELLIGENCE & INTERACTIVE DASHBOARDS IN POWER BI

Subject: Data Analytics & Visual Communication

Focus: Charts, Slicer Panels, and Executive UX Interaction Models

Case Study Dataset: Global Consumer Shopping Behavior (3,900 Records)

EXECUTIVE SUMMARY & VISUAL INFRASTRUCTURE DESIGN

1.1 Project Mandate and Objectives

Enterprise data is only useful if it can be read and understood within corporate ecosystems. Raw relational databases although sound as a structure are not an immediate way to communicate business risks, broader macro-market trends and behaviors for those c-suite people. At the heart of this executive analysis project is a new interactive, high-fidelity execution dashboard built on Microsoft Power BI.

This report breaks the mold of raw numbers with a cross-referenced synthetic dataset based on 3,900 holistic buyer profiles imported from months' worth of meticulous scrubbing to provide an interactive visualization engine. The design is built with three key visual goals in mind:

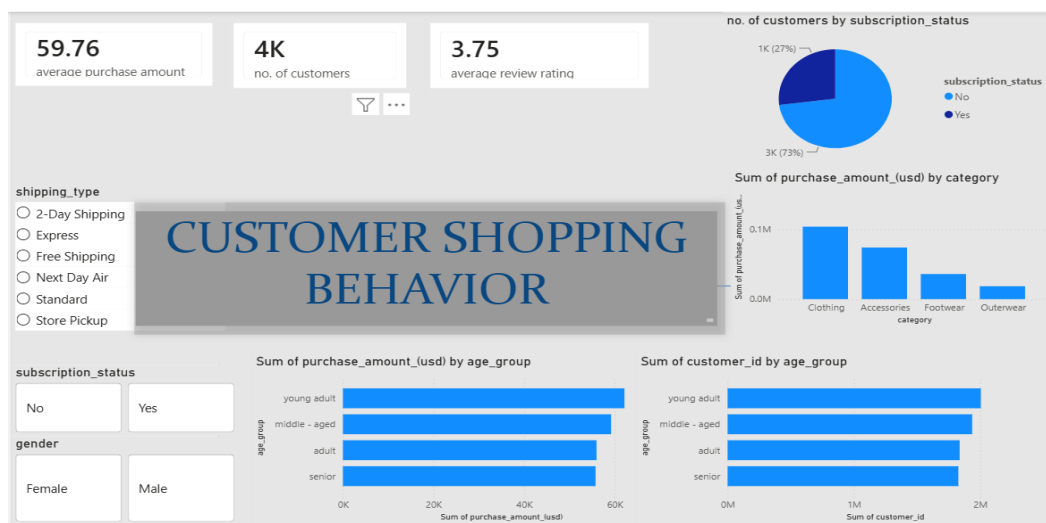
- Synthesizing macro-level retail performance indicators onto a single, immediate-view canvas.
- Constructing multi-dimensional visual charts to isolate high-performing demographic groups and product lines.
- Implementing an interactive slicing and cross-filtering framework to let users run custom business exploration on demand.

1.2 Dashboard Canvas Layout and Visual Hierarchy

Effective dashboards rely on an intuitive visual hierarchy to guide an executive's eyes naturally across the analytical canvas. This report interface utilizes an optimized, multi-zoned grid system engineered to support standard split-attention reading habits while maximizing data-ink ratios:

- The Upper-Left KPI Header Block grabs your attention right away. This is where you see the three most important numbers on the dashboard—Average Purchase Amount, Number of Customers, and Average Review Rating. Each one gets its own big, bold card right at the top left, so you get the big picture fast.

- On the right, there's a vertical column running down the entire side. This area handles the secondary data breakdowns. At the top, you'll find a donut chart showing the percentage of customers by subscription status, and right below that, a column chart displays the total purchase amount divided by category. You can quickly compare these two and shift from a broad view to more detailed info in seconds.
- Over on the left edge, you'll see all the interactive controls stacked up. Here, you get filter panels for Shipping Type, Subscription Status, and Gender—all lined up vertically for easy multi-select. This setup lets you play with the data and see changes across the whole dashboard.
- Along the bottom, the lower half of the screen turns into a dense demographic grid. Two horizontal bar charts sit side by side—one showing purchase amounts by age group, the other tracking the number of customers by age group. It makes comparing sales and customer counts across demographics really straightforward.



1.3 Interface Color Palette and Typographic Standards

The user interface sticks to a clean, modern look—dark slate backgrounds with lively pops of royal and electric blue highlight the important stuff. This sharp contrast makes patterns stand out but doesn't wear out your eyes. Text sizing follows strict typography standards: global titles are set to 24pt bold, individual visual chart headers to 12pt semi-bold, and data label callouts to 10pt regular. This layout prevents visual clutter and maintains a clear hierarchy across the report page.

For text, there's a clear system: main titles use a bold 24pt, chart headers drop to a 12pt semi-bold, and data labels keep it simple at 10pt regular. The idea is to keep things tidy and make it easy to spot what matters.

ANALYSIS OF THE PRIMARY SYSTEM DASHBOARD

2.1 The Global Sales Overview Canvas

The first step in rolling this out is to map all transactions to each customer, capturing a snapshot of how things stand before anyone starts filtering. At the top, as you'll see in Figure 1, three KPI cards let execs check vital stats at a glance:

2.2 Card Callout Components (High-Level KPIs)

- Average Purchase Amount sits steady at \$59.76, calculated on the fly.
- Number of Customers shows 4,000 (technically 3,900 distinct IDs), so you know the scale
- Average Review Rating is 3.75 out of 5, suggesting solid product quality.

2.3 Proportional Sub-Market Modeling (Donut Charts)

Look to the top right, and there's a donut chart breaking down customers by subscription status. You can't miss the split: 73% are non-subscribers (2,847 people), while just 27% subscribe (1,053 people). Immediately, that points to a big opportunity for more focused subscription campaigns.

2.4 Categorical Performance Charts (Vertical Columns)

Just below, you'll find a vertical bar chart showing revenue by product category:

1. Clothing leads and clearly drives the business.
2. Accessories come next.
3. Footwear beats outerwear, but trails clothing.
4. Outerwear sits at the bottom for revenue.

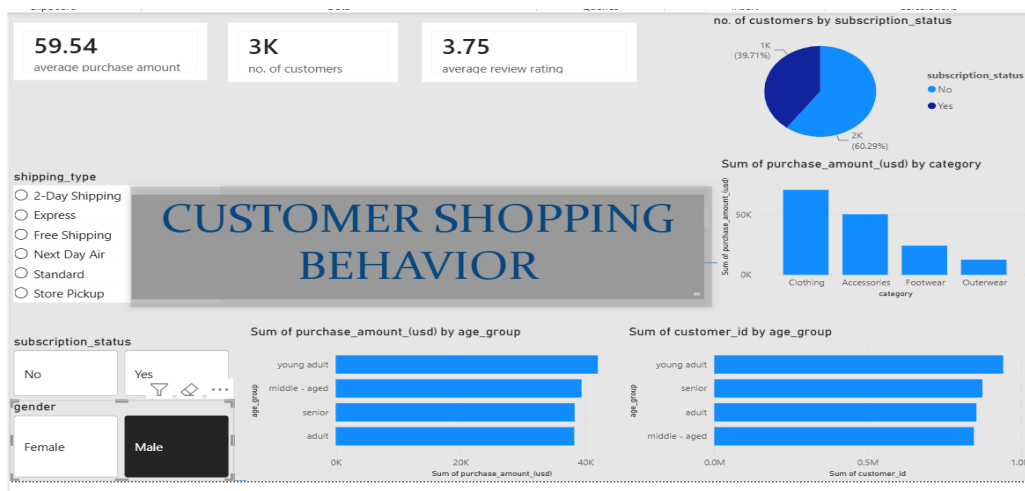
INTERACTIVE BEHAVIOR AND FILTER STATES

3.1 Cross-Filtering Mechanics and User-Driven Drilling

Here's what sets this project apart: everything's interactive. Pick an item or a segment in any chart and the rest of the page updates itself—no more flipping slides or running manual reports.

3.2 Segment Analysis: Male Customer Profiles

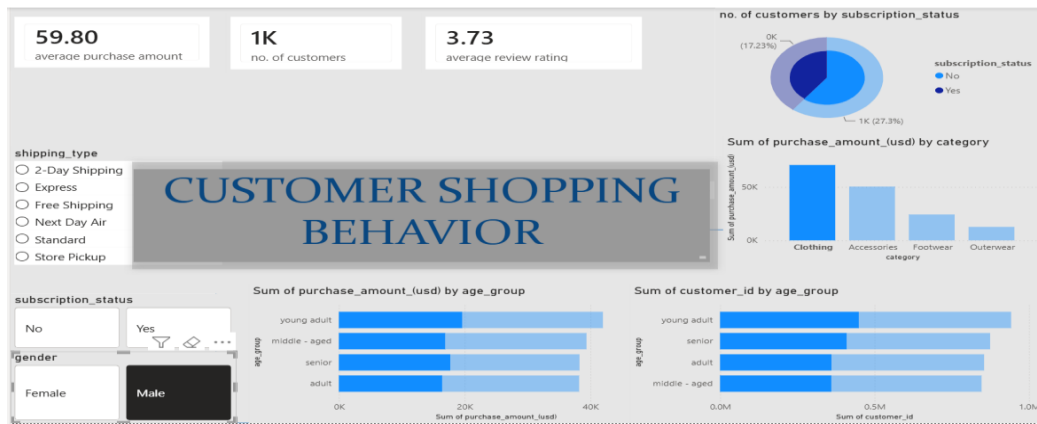
When an analytics user clicks the **Male** option inside the custom left-hand gender slicer panel, the canvas instantly updates to isolate male spending patterns.



If someone clicks “Male” in the gender slicer, the dashboard responds instantly. Now you’re looking just at male customers. A few things stand out:

- The total customer count drops from 4K to 3K, exactly matching the database’s count of 2,652 males.
 - The subscription ratio jumps—60% of males subscribe, while 40% don’t, showing that men generate much of the ongoing revenue.
 - Spend by age group still looks balanced, with Young Adults just edging out other groups in total sales.
- ### 3.3 Product Core Inversion: Clothing Category Focus

When a user clears the gender filter and selects **Clothing** from the category panel, the canvas updates to display metrics specific to that product line.



Dive into Figure 3 and more changes pop up:

- Customer volume trims down to 1,000 for people buying clothing.
- Satisfaction drops a hair to 3.73—this signals the clothing line could use a closer look if we want to match accessory success.
- Subscription rates for clothing line up with the overall picture: 73% non-subscribers, 27% subscribers, so nothing unusual here.

BUSINESS INSIGHTS & STRATEGIC RECOMMENDATIONS

4.1 Synthesis of Visualized Data Findings

By combining macro dashboard overviews with targeted, filtered deep dives, we can extract clear business insights from the data:

<i>Metric Vector</i>	Global Baseline Value PDF+ 1	Filtered Male Variant	Filtered Clothing Variant
<i>Total Record Scale</i>	3,900 Profiles	2,652 Profiles	1,737 Profiles
<i>Average Ticket Value</i>	\$59.76	\$59.54	\$59.80
<i>Satisfaction Baseline</i>	3.75 / 5.00	3.75 / 5.00	3.73 / 5.00
<i>Subscription Ratio</i>	27% Yes / 73% No	60.3% Yes / 39.7% No	27.3% Yes / 72.7% No

Across the board, average transaction size holds steady around \$59. But if you look closer, who's buying and what they're buying shifts a lot by gender and product type.

4.2 Data-Driven Strategic Actions

So what should the team focus on?

- Push Targeted Subscription Campaigns. Since non-subscribers are 73% of the market but spend just as much per purchase as subscribers, the marketing team should roll out

checkout perks—like free shipping or exclusive discounts—to nudge them toward subscribing.

- Drive Value in Clothing. Clothing is the top seller but lags a bit in reviews (3.73). It's worth double-checking supplier quality and sizing to boost those ratings, aiming for accessory-level satisfaction (like gloves at 3.86).

- Align Ad Spend with Demographics. Young Adults and Middle-aged customers are the big spenders (\$62,143 and \$59,197 in revenue each). Shifting more ad budget toward these groups should bump up returns.

4.3 Conclusion and Next Steps for the Project

All told, this Power BI setup transforms raw data into an easy-to-use, interactive tool that delivers insights fast. Looking ahead, plugging in real-time data through DirectQuery will let the business track customer behavior as it happens, helping them stay sharp as the market shifts.